Alexandra Botterly

1. Optimizing composition for F/M steels → can determine amount of Cr to include, which in high amounts can cause δ Ferrite phase to form ~~but too~~ and embrittlement / hardening. Too little and the corrosion resistance isn't optimized

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→ can also add W and Mo at ~3 wt% where >3 wt% it is stronger but more brittle and ↑ n absorption cross section
~~can also tailor~~ C added → increase $M_{23}C_6$, decreased → more MX phases
→ Si can be added, Ni concentration can be tailored for application as well
→ tailor composition so ideal amounts of ferrite vs martensite

10/10

2. Ferritic steels are BCC, which has a less dense packing than ~~FCC~~ Austenitic (FCC), which means they can accommodate more ~~swelling~~ FP within a single unit. The lattice strain also repels other interstitials and absorbs vacancies, preventing them from ~~agglomerating~~ clustering to form voids and cause more swelling.
(I think if more inherent defects like dislocation, those act as sinks. missing a big one though)

voids, not FP 7/10 - got pieces of it...

3. DBTT → Ductile-Brittle Transition Temperature → when a materials means of failure turns from being dductile failure to a brittle failure. BCC alloys experience an increase in DBTT even at low T_{irr} and doses, ~~meaning the T~~ ~~op with water~~ This can lead to abrupt failure of cladding at an unexpected T that the clad is operating at. Irradiation embrittlement is caused by defects formed during irradiation, which may impede upon dislocation motion in the alloy, causing it to be harder, stronger, and more brittle. Martensitic laths ~~are~~ have large networks of dislocations and can absorb defects to mitigate irradiation hardening, but ultimately ppt hardening and irradiation hardening lead to a reduction in Total Elongation and ↑ in YS

8/10

- precipitates, composition effects.

4. FeCrAl alloys have a great corrosion resistance due to Cr and Al within the system, which both form protective passive layers that repair themselves. These thin oxide films prevent further oxidation of the ~~matrix~~ bulk alloy and are stable.

→ Too little Cr and Al results in an incomplete passive film and minimizes corrosion, Too much Cr and Al can lead to embrittlement (or lack of ↑T stability if used in LWR)

8/8

FeCrAl is of interest for LWRs as an ATF to prevent an accident like Fukushima, and also for Advanced reactors

↙ they strengthen!

5. ODS steels → Oxide Dispersion Strengthened Steels

Oxide particles strengthen materials and also act as defect sinks Y_2O_3 is a common oxide used. Oxides can strengthen materials by preventing dislocation motion, ~~and same~~ acting as an obstacle. They also can trap irradiation defects and prevent swelling by creating lattice mismatch for defects to reside within by different particle sizes

6/6

6. Advantages of Ni Alloys → good corrosion resistance in molten salts, maintain favorable mechanical properties up to high Temperatures

8/8

Disadvantages → reduction in swelling resistance due to FCC type lattice, n,α transmutation to form He, which leads to He embrittlement, reduction of hardness with ↑T and ↑dose

Alexandra Doherty

7. USi vs UAl fuels 12/12

USi fuels were introduced after ~~with~~ Reduced Enrichment for RTR by DOE, when DOE mandated the use of Low enriched U in RTRs.

UAl fuels experienced no void swelling, due to the peritectic formation of UAl_3 and UAl_4 by UAl_2 reacting with Al cladding, which results in a reduction of volume, offsetting any swelling. However, this same reaction is the reason why there is a much lower U at% in UAl_x fuels opposed to USi fuels. USi fuel forms are U_3Si, U_3Si_2, and USi, allowing for optimization of fissile at. density. However, USi fuels do swell, a lot more than UAl, and also form an interaction layer (IL) with Al cladding, leading to clad blistering, delamination, and potential clad burst / FP release.

U_3Si_2 is more viscous than U_3Si, meaning it has a slower diffusion leading to less ~~[illegible]~~ agglomeration of defects. U_3Si_2 also has a smaller free V, leaving less free area for FG to reside. U_3Si is not as viscous, leaving fast diffusion for FG to collect in free volume, agglomerate, and swell. U_3Si bubbles are bimodal (large and small) and ~~U_3Si_2~~ unstable, whereas U_3Si_2 has more stable bubble growth & small bubbles ~ equal sized.

8. γ phase is not dominant @ reactor temperatures but is in-reactor due to its inability to make an α-γ phase transformation. PPts (I think) from irrad. pin motion of dislocations and prevent the γ phase from turning to α ✗

4/7 - not really...

9. Solidus-liquidus (sounds like a spell in Harry Potter) gap ~~is what happens of Mo causes~~ wingardia leviosa! 6/6

in UMo fuels, Mo cools fastest and can cause a "segregation" of Mo via Mo islands. UMo is ductile and hard to use fragmentation, but geometry / irregularity of particles is not a huge factor in UMo due to no Heat Treatment. during atomization, the UMo is eventually cooled, but in a cellular fashion with Mo-rich islands and Mo-lean matrix. When the UMo is rolled, these Mo-rich islands elongate and create a disordered grain structure with a

→ back

cotd...

With a non-homogenous Mo distribution. To eliminate this, an additional annealing stage is used to recrystallize grains and homogenize chemical structure. Fragmentation is ~~nua~~ probably better because it is cheaper and SA/V ratio and that doesn't matter as much as in other intermetallic fuel Applications

10. At low BU → there are FG bubbles (small) along GBs but not seen by OM in matrix

→ although shown via TEM, a FG superlattice exists with ~2nm bubbles that aren't enough for a significant ΔV, but proves they are there.

At mid BU → FG bubbles get larger along GB and some form in matrix.

→ Defects from irrad migrate to GBs, which saturate and grain refinement begins.

↳ polygonization of fuel takes place from GBs into matrix, destroying the FG superlattice and absorbing the superlattice bubbles to further ~~g~~ refine grains / restructure the material

At high BU → ~~Fug~~ Grain refinement nears completion until ~~entire grain~~ All fuel has been consumed

(bad drawings, sorry)

10/10

Kh.

Alexandra Boberry

11. Interaction layer of UMo - Al cladding is important because bubbles form along here and eventually lead to delamination and cladding failure. To mitigate this, coated UMo particles, add Si into the system, or use a monolithic instead of dispersed UMo fuel (U-10Mo) with a Zr interdiffusion layer

7/7 → Zr interdiffusion layer is more stable with Al and UMo than they are with eachother. Failure via delam can still happen but not as quickly, and swelling / inward creep is of higher significance than dispersed

Interaction layer also amorphous and causes ↑ in swelling inherently (running out of time)

12. Al has a melting point of ~660°C, which is incredibly low. RTRs are used more for materials test / isotope production and operate at T ~ 100-150°C, which is much lower. Al is very cheap, abundant, ductile, corrosion resistant. Just not high T resistant or very strong, which isn't as important with RTRs

6/6